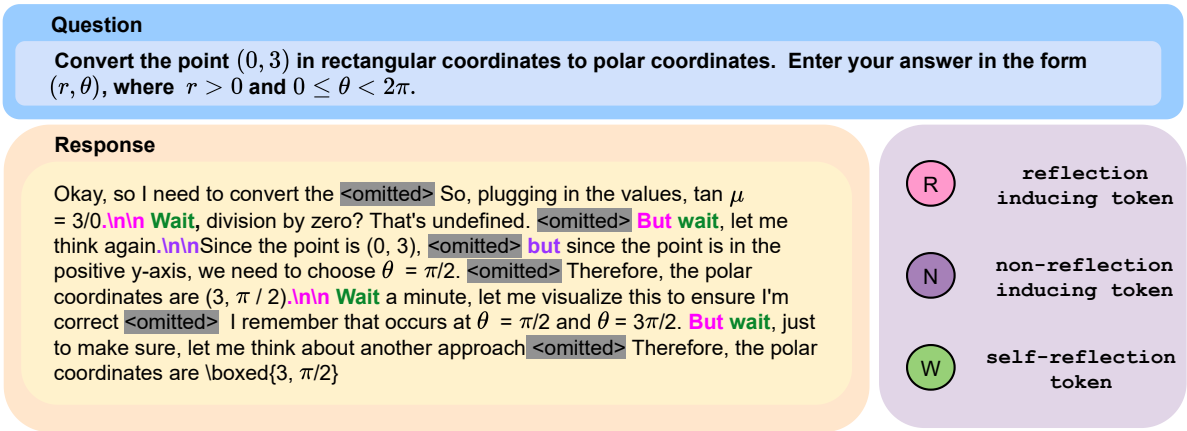

Figure 2: **Hidden State Selection Methodology:** We identify tokens immediately preceding "wait" tokens as reflection-inducing tokens and extract their hidden states. For comparison, we collect hidden states of identical tokens appearing in non-reflective contexts. This contrastive approach enables us to analyze the neural signatures associated with self-reflection in language models.

instances are characterized by explicit reconsideration or revision of prior reasoning steps. For details on how we identify self-reflection instances, please see Appendix B. While the self-reflection tokens differ somewhat from those typically observed in RLVR-trained models, their reflective nature is still discernible. These findings suggest that self-reflection is not solely acquired through fine-tuning, but rather emerges as a latent capability within the $\mathcal{A}_{\texttt{pt}}$—one that is infrequently activated but nonetheless present.

Reflection-Inducing Probing by Injecting CoT from $\mathcal{A}_{\texttt{ft}}$ into $\mathcal{A}_{\texttt{pt}}$ However, the extremely low frequency of self-reflection in A_{pt} makes it challenging to analyze systematically and to develop methods (which will be studied in the next section) for controlling such behavior. To address this challenge, we propose a probing method, termed *reflection-inducing probing*, that isolates the model's capacity for self-reflection from its general reasoning ability. The key idea is to decouple reasoning competence from reflective behavior by inserting reasoning traces generated by the fine-tuned model $\mathcal{A}_{\texttt{ft}}$ into the input of the pretrained model $\mathcal{A}_{\texttt{pt}}$, and then measuring whether $\mathcal{A}_{\texttt{pt}}$ generates reflection in response.

Formally, given a question q, we use the fine-tuned model $\mathcal{A}_{\texttt{ft}}$ to generate a sequence of reasoning tokens:

$$\mathcal{A}_{\texttt{ft}}(q) = (\underbrace{r_1}_{\text{pre-reflection}}, \underbrace{\texttt{reflection}}_{\text{explicit signal}}, \underbrace{r_2}_{\text{post-reflection}})$$

where r_1 denotes the initial chain-of-thought leading up to an explicit reflection token (e.g., "wait"), and r_2 represents the revised or continued reasoning after reflection. We then construct a new prompt by inserting r_1 (the pre-reflection reasoning) into the input of $\mathcal{A}_{\texttt{pt}}$, and evaluate whether $\mathcal{A}_{\texttt{pt}}$ independently produces a reflection token at the appropriate point. This setup ensures that both models operate on similar reasoning contexts, eliminating confounding differences in reasoning capability. By comparing the frequency and consistency of self-reflection under this controlled setting, we can more directly assess whether reflective behavior is present in the pretrained model and to what extent it is amplified by fine-tuning. The frequency of generating reflection with reflection-inducing probing is reported in Figure 1.

Self-Reflection emerges naturally in pretrained models albeit with much lower frequency Remarkably, $\mathcal{A}_{\texttt{pt}}$ exhibits clear self-reflective behavior in 18.6% of these cases—a dramatic increase from its baseline rate. This differential response demonstrates that while $\mathcal{A}_{\texttt{pt}}$ rarely produces overt reflection markers in standard contexts, it possesses latent self-reflection capabilities that can be activated by appropriate contextual triggers. These findings strongly suggest that self-reflection mechanisms are encoded during pretraining, rather than being exclusively developed through reinforcement learning. With 18.6% self-reflection cases, the subsequent

section analyzes the hidden state representations underlying these self-reflective behaviors to provide further insights.

3.2 Hidden State Representations of Self-Reflection

To further investigate the emergence of self-reflection, we analyze the internal representations of the model when it decides to generate reflection versus when it does not. Specifically, we focus on the hidden states associated with reasoning tokens that immediately precede the generation of a reflection token (e.g., "wait"), and compare them to those that do not lead to reflection.

Since both the pretrained model $\mathcal{A}_{\texttt{pt}}$ and the fine-tuned one $\mathcal{A}_{\texttt{ft}}$ exhibit self-reflection behaviors, we use $\mathcal{A}$ to denote a generic model (either pretrained or fine-tuned), which will be specified in context. Given a question q, suppose the model $\mathcal{A}$ generates a sequence of reasoning tokens that includes self-reflection. Let $r = \mathcal{A}(q) = (r_1, \texttt{reflection}, r_2)$, where r_1 precedes the reflection token and r_2 follows it. Due to the auto-regressive nature of transformer models, the information from the question q and the reasoning tokens r_1 is aggregated into the hidden representation of the final token in r_1, which is then used by the last layer to predict the next token—the reflection token. For convenience, we refer to the final token in r_1 as a *reflection-inducing token*, though its hidden state captures information from the entire preceding context (q, r_1). Reflection-inducing tokens often coincide with sentence-final punctuation (e.g., ".", "!", or closing brackets) or specific markers such as "But". Now with a slight abuse of notation, let $h^{(\ell)}_{\text{reflection-inducing}}(q, r)$ denote the ℓ-th layer hidden state of a model at $\mathcal{A}$ corresponding to the reflection-inducing token. We collect all such hidden states from model outputs that contain reflection tokens into the following set

$$\mathbb{H}^{(\ell)}_{\texttt{reflect}} = \left\{ h^{(\ell)}_{\text{reflection-inducing}}(q, r) \in \mathbb{R}^d \right\}. \tag{2}$$

To study the properties of hidden states associated with self-reflection, we contrast this set with representations from cases where the model *does not* generate reflection. Specifically, to eliminate the confounding effect of token surface form, we extract hidden states from tokens that share the same form as reflection-inducing tokens (e.g., sentence-final punctuation), but which do not lead to self-reflection in the subsequent responses (within 100 tokens in the experiments). For notational convenience, we refer to these as *non-reflection-inducing tokens*, and denote their corresponding hidden states as $h^{(\ell)}_{\text{non-reflection-inducing}}(q, r)$. See Figure 2 for an illustration comparing reflection-inducing and non-reflection-inducing tokens. We collect such non-reflection-inducing into the following set

$$\mathbb{H}^{(\ell)}_{\texttt{non-reflect}} = \left\{ h^{(\ell)}_{\text{non-reflection-inducing}}(q, r) \in \mathbb{R}^d \right\}. \tag{3}$$

This design ensures a fair comparison by controlling for the surface form of the reflection-inducing token, ensuring that any differences in hidden representations are attributable to the model's decision to reflect.

For $\mathcal{A}_{\texttt{pt}}$ models, which rarely generate self-reflective outputs, we use the method of reflection-inducing probing by injecting CoT from $\mathcal{A}_{\texttt{pt}}$ into $\mathcal{A}_{\texttt{ft}}$ to elicit reflective behavior. To visualize these high-dimensional representations, we employ UMAP dimensionality reduction [31], projecting the hidden states into a 2D space. Figure 3 presents the visualizations for the 15th layer (out of 28 total layers) for both models, with more layers presented in Appendix A.

Our analysis reveals a pattern: **both models show clear separation between self-reflection and nonself-reflection states**. While this separation is expected in $\mathcal{A}_{\texttt{ft}}$, which was explicitly trained to exhibit self-reflective behavior, the equally distinct clustering in $\mathcal{A}_{\texttt{pt}}$ is remarkable. Despite rarely generating explicit self-reflection tokens in its outputs, $\mathcal{A}_{\texttt{pt}}$ maintains internal representations that clearly distinguish between self-reflective and nonself-reflective contexts. This finding provides strong evidence that self-reflection capabilities develop during pretraining, with models encoding these patterns in their hidden state representations even when they rarely manifest in generated text. We will exploit this internal structure to develop methods for controlling self-reflection in the next section.

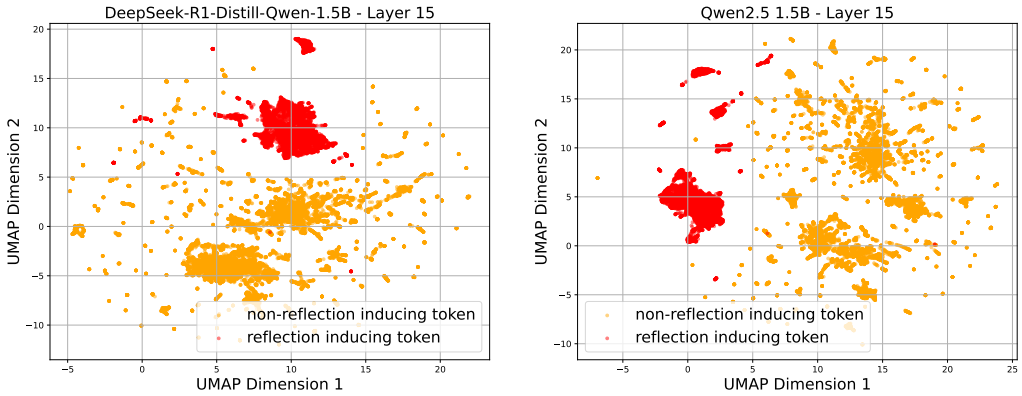

Figure 3: **UMAP visualization of hidden state representations** of (Left) $\mathcal{A}_{\texttt{ft}}$ and (Right) $\mathcal{A}_{\texttt{pt}}$. Both models show separation between $\mathbb{H}^{(\ell)}_{\texttt{reflect}}$ and $\mathbb{H}^{(\ell)}_{\texttt{non-reflect}}$.

4 Controlling Self-Reflection in Language Models

In this section, we introduce our approach for identifying and manipulating self-reflection vectors in LLMs. Building on our finding in the last section that hidden representations distinctly separate self-reflective from non-reflective contexts, we construct self-reflection vectors, directions in activation space associated with self-reflective reasoning. We then demonstrate how these vectors can be used to bidirectionally control self-reflection behavior, either enhancing it to improve reasoning accuracy or suppressing it to reduce computational overhead. Through extensive evaluation across multiple mathematical reasoning benchmarks, we show that our method significantly outperforms strong baselines while offering flexible control over the performance-efficiency trade-off. Finally, we examine the cross-domain transferability of these vectors, revealing their potential as universal controls for self-reflection across diverse reasoning tasks.

4.1 Extract Self-Reflection Vectors

To identify the self-reflection vector in the residual stream activations, we compute the difference between the activations of self-reflective and nonself-reflective contexts. This technique, known as difference-in-means, effectively isolates key feature directions, as demonstrated in prior work [16, 32, 33], motivating our application of this approach to the self-reflection domain. As in Section 3, we focus on the hidden state of reflection-inducing token, positing that this state encodes the model's transition into self-reflection reasoning.

For each layer $\ell \in \{1, \ldots, L\}$, we compute the mean hidden states in the reflection set $\mathbb{H}^{(\ell)}_{\texttt{reflect}}$ and non-reflection set $\mathbb{H}^{(\ell)}_{\texttt{non-reflect}}$ as

$$\boldsymbol{\mu}^{(\ell)}_{\texttt{reflect}} = \text{mean}(\mathbb{H}^{(\ell)}_{\texttt{reflect}}), \quad \boldsymbol{\mu}^{(\ell)}_{\texttt{non-reflect}} = \text{mean}(\mathbb{H}^{(\ell)}_{\texttt{non-reflect}}), \tag{4}$$

and then construct the self-reflection vector as the difference-in-means vector

$$\boldsymbol{v}^{(\ell)} = \boldsymbol{\mu}^{(\ell)}_{\texttt{reflect}} - \boldsymbol{\mu}^{(\ell)}_{\texttt{non-reflect}}, \tag{5}$$

which captures both the direction along which self-reflective and nonself-reflective activations diverge, and the magnitude of that divergence.

4.2 Model Interventions for Controlling Trade-off between Reasoning and Efficiency

To actively modulate a model's tendency to reflect, motivated by the linear representation hypothesis and prior work [32], we apply simple linear interventions based on the self-reflection vector $\boldsymbol{v}^{(\ell)}$ extracted from the ℓ-th